VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590018

A MAJOR PROJECT REPORT (BCSP786) ON

"SMART MEDICINE MANAGEMENT SYSTEM USING AI & OCR"

Submitted in Partial fulfillment of the Requirements for the Degree of

Bachelor of Engineering in Computer Science &Engineering

By

Madhuri Shah (1CR22CS095)

Shreya Goenka (1CR22CS185)

Swagata Ghosh (1CR22CS198)

Under the Guidance of

Rajni Tiwari

Asst. Professor, Dept. of CSE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI, BANGALORE-560037

CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI,
BANGALORE-560037

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

Certified that the project work entitled "Smart Medicine Management System Using AI and OCR" carried out by Madhuri Shah (1CR22CS095), Swagata Ghosh (1CR22CS198), Shreya Goenka (1CR22CS185) bonafide students of CMR Institute of Technology, in partial fulfillment for the award of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belgaum during the year 2025-2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library.

The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

Rajni Tiwari	Dr.R Kesavamoorthy	Dr. Sanjay Jain
Asst. Professor	Professor & Head	Principal
Dept. of CSE, CMRIT	Dept. of CSE, CMRIT	CMRIT

DECLARATION

We, the students of Computer Science and Engineering, CMR Institute of Technology, Bangalore declare that the work entitled "Smart Medicine Management System Using AI and OCR " has been successfully completed under the guidance of Prof. Rajni Tiwari , Computer Science and Engineering Department, CMR Institute of Technology, Bangalore. This dissertation work is submitted in partial fulfillment of the requirements for the award of Degree of Bachelor of Engineering in Computer Science and Engineering during the academic year 2025 - 2026. Further, the matter embodied in the project report has not been submitted previously by anybody for the award of any degree or diploma to any university.

Place: Bangalore

Date: 22/11/2025

Team members:	Signature
Madhuri Shah (1CR22CS095)	_____
Swagata Ghosh (1CR22CS198)	_____
Shreya Goenka (1CR22CS185)	_____

ABSTRACT

This project presents a Smart Medicine Management System that integrates artificial intelligence and optical character recognition technologies to automate pharmacy operations. The system provides a dual-portal web application for sellers and buyers, enabling efficient inventory management and seamless medicine ordering. Google Vision API and Tesseract OCR are used for extracting text from medicine strips and handwritten prescriptions. BioBERT-based natural language processing is applied to interpret and explain medical prescriptions for better user understanding. The system supports real-time stock updates, expiry alerts, and alternative medicine suggestions. It reduces manual workload, minimizes human errors, and improves accuracy in medicine handling. The platform enhances communication between pharmacies and customers through intelligent automation. Overall, the system delivers a faster, safer, and more reliable digital pharmacy experience.

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude and respect to **CMR Institute of Technology, Bengaluru** for providing me a platform to pursue my studies and carry out my final year project

I have a great pleasure in expressing my deep sense of gratitude to **Dr. Sanjay Jain,** Principal, CMRIT, Bangalore, for his constant encouragement.

I would like to thank **Dr.R Kesavamoorthy,** Professor and Head, Department of Computer Science and Engineering, CMRIT, Bangalore, who has been a constant support and encouragement throughout the course of this project.

I consider it a privilege and honor to express my sincere gratitude to my guide **Rajni Tiwari, Assist. Professor,** Department of Computer Science and Engineering, for the valuable guidance throughout the tenure of this review.

I also extend my thanks to all the faculty of Computer Science and Engineering who directly or indirectly encouraged me.

Finally, I would like to thank my parents and friends for all their moral support they have given me during the completion of this work.

TABLE OF CONTENTS

Certificate ii

Declaration iii

Abstract iv

Acknowledgement v

Table of contents vi

List of Figures ix

List of Tables x

1. INTRODUCTION 11

1.1 Existing System 11

1.2 Drawbacks of the Existing System 11

1.3 Proposed Solution 12

1.4 Objectives of the proposed System 12

2. LITERATURE SURVEY 13

2.1 Domain-Specific Named Entity Recognition (NER) in Healthcare 13

2.2 Techniques for Optical Character Recognition (OCR) of Handwritten Medical Forms 13

2.3 Integration of Large Language Models (LLMs) for Medical Knowledge Retrieval 14

2.4 Architecture for Real-Time Notification and Alert Systems 14

2.5 Performance Metrics and Comparative Evaluation of NER Models 15

3 System Requirements and Design Specifications 19

3.1 Minimum Hardware Requirements 19

3.2 Minimum Software Requirements 19

3.3 Platform and Hosting Requirements 20

3.4 System Architecture 20

3.5 Functional Requirements 21

3.6 Non- Functional Requirements 21

3.7 Module Overview 21

3.8 System Constraints 22

4 METHODOLOGY 23

4.1 System Workflow 23

4.2 Methodological Approach 24

4.3 Data Flow Description 24

4.4 Sequence of Prescription Analysis Execution 25

4.5 Activity & Sequence Diagrams 26

4.6 Algorithmic Steps 27

4.7 Summary 27

5 IMPLEMENTATION 28

5.1 Implementation Overview 28

5.2 Development Tools 28

5.3 Technology Stack 29

5.4 Implementation Details 30

5.5 Code Structure & Package Layout 31

5.6 Key Implementation Snippets 31

5.7 Testing and Debugging Strategy 32

5.8 Summary 32

6 RESULT AND DISCUSSION 33

6.1 Experimental Setup 33

6.2 Module-Wise Output Evaluation 34

6.3 Integrated Workflow Results 34

6.4 System Performance Evaluation 35

6.5 Screenshots 36

6.6 Result Analysis **38**

6.7 Summary **39**

7 TESTING **40**

7.1 Conclusion **40**

7.2 Limitations **40**

7.3 Future Scope **41**

7.4 Summary **42**

8 CONCLUSION **43**

8.1 Summary of Work Carried Out **43**

8.2 Contributions of the Project **43**

8.3 Utility of the System **44**

8.4 Scope for Future Work **44**

8.5 Summary **45**

REFERENCES **46**

LIST OF FIGURES

	Page No.
Fig 4.1 Activity Diagram for Overview of MedsAI Prescription Analysis Workflow	**26**
Fig 4.2 Sequence Diagram of Hybrid of MedsAI Prescription Analysis	**26**
Fig 6.1 Chatbot Interface	**36**
Fig 6.2 Prescription Image Upload Screen	**37**
Fig 6.3 AI Result Detailed Medical Explanation	**37**
Fig 6.4 Seller Dashboard: Inventory and Stock Alerts	**38**

LIST OF TABLES

Page No.

Table 2.1 Comparison Table **16**

CHAPTER 1

1. INTRODUCTION

The healthcare and pharmaceutical industry is rapidly evolving with the integration of digital technologies to improve efficiency, accuracy, and accessibility. One of the major challenges faced by pharmacies and patients is the manual management of prescriptions and medicine inventory, which often leads to errors, delays, and poor tracking of critical information such as expiry dates and stock availability. With the rise of Artificial Intelligence (AI) and Optical Character Recognition (OCR), it has become possible to automate and streamline these processes.This project falls under the domain of Healthcare Information Systems and Intelligent Pharmacy Management, where modern web technologies and AI models are used to assist both medicine sellers and buyers. The system bridges the gap between handwritten prescriptions and digital pharmacy operations by converting prescription images into meaningful, structured information and providing intelligent recommendations.The proposed application is a dual-portal web platform that supports sellers in managing stock efficiently and helps buyers understand prescriptions, track medicines, and place orders easily. By integrating advanced tools such as Google Vision OCR and BioBERT-based Natural Language Processing (NLP), the system ensures greater accuracy, speed, and safety in medicine handling.

1.1.Existing System

The traditional pharmacy system mostly relies on manual processes for handling handwritten prescriptions, managing medicine inventory, and tracking sales. Pharmacists manually enter stock details, verify prescriptions by visually interpreting handwriting, and maintain sales records either in paper registers or basic software systems. Customers often struggle to understand medical prescriptions and depend entirely on pharmacists for correct medicine selection.

1.2.Drawbacks of the Existing System

The existing manual and semi-digital systems are prone to human errors such as incorrect medicine dispensing, wrong dosage interpretation, and delayed stock updates. There is no real-time synchronization between prescription understanding and inventory availability. Customers face difficulties in tracking their medicines, checking availability, or finding alternatives. Additionally, expiry management is often neglected, leading to wastage and potential health risks due to expired medicines remaining in stock.

1.3.Proposed Solution

The solution is to develop an AI-powered Smart Medicine Management System that automates prescription reading, inventory management, and medicine ordering. The system will use Google Vision OCR to extract text from medicine strips and handwritten prescriptions, while BioBERT NLP will analyze and explain prescription details. The integrated dual-portal system will allow sellers to manage stock in real time and buyers to track medicines, receive alerts, and place orders based on accurate and updated inventory data.

1.4. Objectives of the Proposed System

- To design and develop a dual-portal web application with role-based features for sellers and buyers.
- To automate prescription and medicine information extraction using OCR and NLP to minimize human errors.
- To implement real-time inventory management, expiry alerts, and intelligent brand alternative suggestions.
- To enable seamless medicine ordering while improving overall efficiency, accuracy, and transparency in pharmacy operations.

CHAPTER 2

LITERATURE SURVEY

The development of the **MedsAI** web application leverages recent advancements in computer vision, deep learning for natural language processing, and medical informatics. Research over the past few years has focused on moving healthcare documentation from paper to digital platforms, emphasizing the need for high-accuracy automated systems capable of reading diverse medical data, including prescriptions and labels. This survey reviews established techniques crucial to creating the Hybrid AI Pipeline, specifically focusing on data extraction and domain-specific knowledge retrieval, which form the foundation of MedsAI's intelligent functionality.

2.1 Domain-Specific Named Entity Recognition (NER) in Healthcare

A major focus area in medical NLP research is the identification and classification of specialized terms within clinical text, known as Named Entity Recognition (NER). In healthcare, this involves accurately tagging entities such as drug names, dosages, frequencies, and medical conditions. Traditional NER methods relied on dictionaries and rule-based systems, but these approaches struggle with variations, misspellings, and novel terminology. Recent studies highlight the superior performance of **Transformer-based models**, specifically those pre-trained on vast amounts of biomedical literature, such as PubMed and clinical notes. These models, exemplified by BioBERT, excel at understanding the context of medical language, significantly boosting the accuracy of extraction tasks critical for patient safety, such as automatically identifying the correct medicine name from a doctor's prescription

2.2 Techniques for Optical Character Recognition (OCR) of Handwritten Medical Forms

- To design and develop a dual-portal web application with role-based features for sellers and buyers.
- To automate prescription and medicine information extraction using OCR and NLP to minimize human errors.
- To implement real-time inventory management, expiry alerts, and intelligent brand alternative suggestions.

- To enable seamless medicine ordering while improving overall efficiency, accuracy, and transparency in pharmacy operations.

2.3 Integration of Large Language Models (LLMs) for Medical Knowledge Retrieval

Beyond simple extraction, modern healthcare systems utilize Large Language Models (LLMs) to transform raw clinical data into actionable, patient-friendly information. Research shows that while specialized BERT models handle precise extraction (NER), general LLMs are ideal for **Generative Question Answering** and synthesizing complex information. Techniques involve conditioning the LLM with the highly accurate data extracted by the NER model (e.g., drug name and dosage) and prompting it to generate explanations regarding use, composition, and potential side effects. This two-stage, or **Hybrid Approach**, ensures that the system maintains high factual integrity through the specialized extractor while providing rich, comprehensive, and easily digestible explanatory content, enhancing patient understanding and medication adherence

2.4 Architecture for Real-Time Notification and Alert Systems

Reliable notification architecture is crucial for patient adherence trackers and inventory management. Studies on real-time alerting focus on high-reliability, low-latency messaging frameworks. Research in this area supports the use of decoupled architectures, often involving asynchronous task queues (like Celery) and dedicated messaging protocols (like Firebase Cloud Messaging for mobile push) to ensure alerts are delivered consistently, regardless of the primary web application's load. For inventory, the literature recommends event-driven triggers based on database actions (e.g., checking stock or expiry fields upon a record save) rather than periodic polling, ensuring instantaneous response to critical changes in stock levels or expiry dates

2.5 Performance Metrics and Comparative Evaluation of NER Models

A significant body of work focuses on the rigorous evaluation of specialized deep learning models in healthcare. The consensus emphasizes using metrics beyond simple accuracy. Literature consistently highlights the necessity of using the **F1-Score** as the primary measure, as it provides a balanced assessment of both **Precision** (avoiding false drug identifications) and **Recall** (avoiding missed drugs). Comparative studies across various benchmarks (e.g., BC5CDR) routinely show that specialized models like BioBERT significantly outperform general-purpose LLMs in sequence labeling tasks, validating the choice of domain-specific architecture for high-stakes medical applications

S.No	Technique Reviewed	AI Methods Used	Application Domain	Key Contributions	Dataset / Validation
1	NLP & Patient Similarity Modeling	Sent2Vec, BioSentVec, ML & DL with NLP	Clinical Decision Support	Symptom-to-disease mapping using similarity in symptom vectors	MIMIC-III ICU Dataset [1]
2	Data Preprocessing & Data-Centric AI	Feature Engineering, Data Augmentation	General AI Systems	Focus on improving data quality for enhanced model robustness and fairness	Multi-domain Applications [2]
3	Blind Image Quality Assessment	RNN + Self-Attention	Image Processing (IQA)	End-to-end quality scoring without reference images; strong correlation with human scores	Benchmark IQA Datasets [3]
4	Change Detection with Siamese U-Net	Siamese U-Net, Attention Modules (ASPP, FAM)	Remote Sensing	Detecting small, noisy changes using multi-scale attention mechanisms	Remote Sensing Datasets [4]
5	Chronic Disease Prediction with Preprocessing	RF, CatBoost, Fuzzy (Sugeno, Mamdani), DL	Chronic Disease Diagnosis	Importance of preprocessing: handling missing data, normalization, feature selection	Various Disease Datasets [5]

6	Validation of ML Algorithms in Primary Care	ML/DL across lifecycle phases	Primary Care Decision Support	Evaluates evidence availability across development, validation, and integration phases	43 Algorithms across AIPA/FDA/CE [6]
7	Health Information Technology (HIT) & EHR Adoption	Not model-focused; HIT architecture & interoperability	Healthcare Systems	Impact of EHR, e-prescribing, and federal initiatives on patient care and data access	National Surveys & Databases [7]
8	AI to Address Health Disparities	Data-Driven ML (Social, Genetic, Environmental Data)	Health Equity	Emphasizes inclusive datasets and ethical oversight to reduce AI bias	Large, diverse population datasets [8]
9	ML Implementation in Healthcare	Systematic Review, CFIR Framework	Clinical ML Integration	Identifies implementation barriers/facilitators in real-world ML adoption	34 Studies from 2013–2023 (PRISMA) [9]
10	AI in Diagnosis, Monitoring & Treatment	Supervised, Unsupervised, Reinforcement Learning	Diagnosis, Monitoring, IoT	Combines learning techniques and IoT for remote care and personalized treatment	Multi-study validation [10]
11	Adaptive Stacking Ensemble for Diabetes Prediction	Stacking (SVM, MLP, XGBoost, CatBoost), SHAP	Diabetes Risk Prediction	Error-ambiguity decomposition, classifier optimization, interpretability via SHAP	BRFSS2015 Dataset (70,691 samples) [11]

Table 2.1 Comparison Table

Research Gap

1. **Real-time Performance**
   - Existing systems struggle with achieving low-latency responses when processing large datasets in real-time.
   - The absence of optimized algorithms limits their applicability in time-sensitive applications.
2. **Scalability**
   - Many solutions fail to maintain consistent performance as the dataset size or number of users increases.
   - Current models lack efficient distributed processing techniques, leading to bottlenecks in high-demand scenarios.
3. **Computational Efficiency**
   - There is a need for more resource-efficient architectures to reduce computation.
   - Existing methods consume high memory and processing power, making them impractical for low-power or edge devices.

CHAPTER 3

System Requirements and Design Specifications

(ML-Based Full-Stack Web Application)

MedsAI is a machine learning-based full-stack web application designed to automate pharmaceutical management and provide intelligent health assistance. The system's architecture is built on a robust server-side framework (**Django**), integrating specialized AI models (**BioBERT, LLM/QA**) for critical processing tasks. This chapter outlines all essential system requirements—hardware, software, platform dependencies, architecture, functional needs, and module specifications. These details ensure the high accuracy, low latency, and reliability required for the medical domain.

MedsAI relies on the server's ability to handle concurrent requests, execute complex model inference, and maintain secure database connectivity. The specifications below ensure seamless operation of all core modules, including OCR processing, BioBERT NER, real-time alerting, and inventory lookups.

3.1 Minimum Hardware Requirements (Server Side)

To ensure reliable performance, high-speed model inference, and concurrent user handling, the following minimum server specifications are recommended:

- **Processor:** Multi-core (8+ cores) 2.5 GHz or higher for stable concurrent request handling and parallel AI model inference.
- **RAM:** Minimum 32 GB RAM to support model loading (BioBERT and local QA pipeline) and maintain fast database caching.
- **Storage:** At least 500 GB SSD storage is mandatory for fast I/O, quick database lookups, and efficient model loading.
- **GPU (Optional but Recommended):** Dedicated GPU (e.g., NVIDIA with 8GB VRAM) for accelerating BioBERT model inference, significantly reducing latency.
- **Network:** High-speed Gigabit Ethernet connection for low-latency client-server communication.
- **Backup:** Reliable backup system for database integrity and disaster recovery.

3.2 Minimum Software Requirements (Server Side)

The application requires the following software environment for execution and development:

- **Operating System:** Linux (Ubuntu 20.04 LTS or later) or Windows Server (for production environment).
- **Python Runtime:** Python 3.10 or higher.
- **Web Framework:** Django 4.x or later.

- **Database Engine:** PostgreSQL 14 or later.
- **Model Libraries:** PyTorch/TensorFlow, Hugging Face Transformers.
- **Task Queue:** Celery or equivalent for asynchronous task management (alerts, reports).
- **Web Server:** Gunicorn/uWSGI (for application) and Nginx/Apache (as reverse proxy).

3.3 Platform and Hosting Requirements

MedsAI is a full-stack application requiring secure server hosting for both the application logic and the database.

- **Development Platform:** Visual Studio Code / PyCharm
- **Programming Language:** Python
- **Web Framework:** Django
- **Design Tools:** HTML/CSS/JavaScript
- **Build System:** pip/virtualenv/Docker (for containerization)
- **Hosting:** Cloud VPS (AWS EC2, Azure VM, Google Cloud Compute Engine) or dedicated server with secure firewall configuration.
- **External Dependencies:** Access to **Google Cloud Vision API** (or similar OCR service) for production-level image-to-text conversion.

3.4 System Architecture

MedsAI follows a **three-tier architecture** with a clear separation between the presentation, application, and data layers, heavily augmented by a specialized **AI Services Layer**.

1. **Presentation Layer (Frontend):**
    - Handles the user interface (UI/UX) for both the Buyer and Seller portals.
    - Built using HTML templates, CSS, and JavaScript.
    - Manages user input, image uploads, and displays real-time chatbot and inventory results.
2. **Application Layer (Django Backend):**
    - **Core Logic:** Handles authentication, routing, and business logic for inventory and sales.
    - **AI Orchestration:** Manages the sequential calls to the OCR, BioBERT, and LLM services.
    - **Alerting & Workers:** Manages asynchronous tasks via Celery (e.g., sending dosage alerts and expiry notifications).
3. **Data Layer (PostgreSQL):**
    - Stores all persistent data: Inventory details, User accounts, Tracker entries, and Coin rewards.
    - Optimized for fast lookups based on extracted medicine names.

4. **AI Services Layer:**
    - **OCR Module:** Executes API calls for image-to-text conversion.
    - **BioBERT Module:** Loaded persistently in memory for low-latency **Named Entity Recognition (NER)**.
    - **LLM/QA Module:** Executes knowledge generation and generic Q&A tasks.

3.5 Functional Requirements

The mandatory functionalities of MedsAI include:

- **F1:** Accurately extract medicine names and dosages from prescription images using the **Hybrid OCR + BioBERT Pipeline**.
- **F2:** Automatically generate a detailed medical explanation (use, composition, side effects) for the extracted drug.
- **F3:** Search and display real-time inventory stock and pricing based on the extracted drug name.
- **F4:** Implement automated, event-driven alerts for **low inventory stock** ($\lt 50$ units).
- **F5:** Implement automated alerts for **near-expiry dates** ($\lt 30$ days).
- **F6:** Provide a functional **Medicine Intake Tracker** with scheduled dosage push notifications.
- **F7:** Allow sellers to manage inventory via OCR-assisted data entry and view automated sales reports.

3.6 Non-Functional Requirements

Important quality attributes include:

- **NFR1 – Reliability:** The system must ensure that the BioBERT NER component consistently maintains an **F1-Score above 90%**.
- **NFR2 – Performance:** The total end-to-end processing time for prescription analysis (latency) must be under **2.0 seconds**.
- **NFR3 – Security:** All sensitive health and inventory data must be protected using modern encryption standards (SSL/TLS, hashed passwords).
- **NFR4 – Scalability:** The server architecture must be capable of handling concurrent requests from both hundreds of buyers and multiple sellers without performance degradation.
- **NFR5 – Maintainability:** The code must be modular (separate buyers and sellers apps) and follow standard Django conventions for easy future updates.

3.7 Module Overview

MedsAI is organized into five core modules corresponding to the dual-portal functionality:

1. **AI Orchestration Module:**
    - Manages the sequential execution of OCR, BioBERT NER, and LLM/QA pipeline calls. Ensures error handling at each AI stage.
2. **Inventory Management Module:**
    - Handles all CRUD operations for the Medicine and Inventory database models. Implements the OCR-assisted data input interface for sellers.
3. **Alerting & Reporting Module:**
    - Contains the asynchronous worker logic (Celery tasks) responsible for monitoring stock levels, checking expiry dates, and generating automated sales reports.
4. **Hybrid Chatbot Module:**
    - Handles all buyer interactions, including generic Q&A and the display of the final prescription analysis results and inventory lists.
5. **Health Tracking Module:**
    - Manages the TrackerEntry and CoinAccount models. Contains the logic for scheduled dosage notification delivery and processing the coin reward system.

3.8 System Constraints and Assumptions

System Constraints

- **OCR Dependency:** The system's accuracy is initially constrained by the quality of the input image (e.g., poor handwriting or photo quality), directly impacting the upstream NER performance.
- **Model Resource:** BioBERT inference requires significant server RAM/VRAM; simultaneous high traffic could strain resources without proper load balancing.
- **API Costs:** Reliance on commercial OCR APIs (e.g., Cloud Vision) introduces a per-transaction cost constraint in a production environment.

Assumptions and Dependencies

- **Dependencies:** Requires the external **Google Cloud Vision API** for image-to-text transcription and reliable access to the Hugging Face **Transformers** library.
- **Data Quality:** Assumes the BioBERT model has been successfully fine-tuned on a high-quality medical dataset.
- **Network:** Assumes the client devices (buyers) have sufficient internet connectivity to receive the complex JSON responses from the server and push notifications.

CHAPTER 4

METHODOLOGY

The methodology adopted for developing the **MedsAI** application focuses on a **Hybrid Artificial Intelligence Architecture** to ensure high accuracy in reading prescriptions and reliable interaction with users. The approach systematically combines specialized machine learning models (OCR and BioBERT) for critical data extraction with large language models (LLM) for personalized content generation. This chapter explains the systematic approach used for identifying requirements, designing core processes, implementing functional modules, and integrating the AI components into a unified Buyer-Seller web framework.

The use of **Python (Django)** enables an **Event-Driven, Modular Architecture** where each AI task is an independent service, enhancing overall reliability and enabling seamless future updates.

4.1 System Workflow

The MedsAI workflow is designed around two main trigger points: Seller Inventory Entry (via OCR) and Buyer SOS/Prescription Upload (via BioBERT/LLM).

4.1.1 Seller Workflow (Inventory Management)

Input: Pharmacist uploads an image of the medicine strip or label.

OCR Processing: Cloud Vision API extracts raw text (Name, Expiry, Composition, etc.).

Database Validation: Python backend processes the text and enters structured data into PostgreSQL.

Alerts & Tracking: Automatic triggers are set for Low Stock ($\lt 50$) and Near Expiry ($\lt 30$ days) monitoring.

4.1.2 Buyer Workflow (Hybrid AI Response)

1. **Trigger Detection:** Buyer uploads a prescription image to the Chatbot.
2. **OCR + BioBERT Execution:**
3. OCR extracts raw text from
4. the prescription.
5. The raw text is fed into the **BioBERT model** fine-tuned for NER.
6. **NER Output:** BioBERT accurately extracts **Drug Name, Dosage, and Frequency**.
7. **LLM/QA Processing:** The extracted entities are passed to the LLM module to generate a detailed, human-readable **medical explanation** (use, composition, side effects).
8. **Inventory Search:** Extracted drug names are used to query the Seller's PostgreSQL inventory.
9. **Response:** The Chatbot presents the medical explanation alongside the list of local pharmacies with available stock and prices.

4.2 Methodological Approach

The methodology combines the MVT (Model-View-Template) Architecture of Django with a Sequential AI Pipeline for data processing.

1. Event-Driven Architecture: The system reacts instantly to API calls (image upload) and database events (low stock trigger), minimizing processing delays.
2. Modular AI Integration: Each AI component is implemented as a separate module, allowing the system to use the right tool for the job:
    - OCR: Handles raw image-to-text conversion.
    - BioBERT: Handles high-accuracy Named Entity Recognition (NER).
    - LLM (Local QA Pipeline): Handles Question Answering and Generative Explanation.
3. Data Persistence: PostgreSQL is used to ensure all inventory records, user history, and stock alerts are persistent, reliable, and quickly queryable.
4. Priority-Based Execution: In the Buyer workflow, the BioBERT NER must execute successfully before the LLM Explanation and Inventory Search can proceed, ensuring data accuracy is prioritized over speed.

4.3 Data Flow Description

The system processes critical medical and inventory data across its various components.

Data Flow Steps:

1. Input Layer: Prescription image or medicine strip image is captured via the web application.
2. Extraction Layer:
    - OCR extracts raw text.
    - BioBERT NER processes raw text into structured entities (Drug: X, Dosage: Y).
3. Processing Layer: The Django Backend validates the entities and performs the inventory lookup (Medicine.objects.filter(...)).
4. Generation Layer: The LLM/QA Model uses the structured entities to generate the detailed medical explanation.
5. Storage Layer: Inventory data is saved/updated in PostgreSQL. User history, intake schedules, and coin counts are saved in PostgreSQL.
6. Output Layer:
    - Buyer: Receives the Chatbot response (Explanation + Inventory List).
    - Seller: Receives the Low Stock/Near Expiry email alert.

4.4 Sequence of Prescription Analysis Execution

Below is the exact sequence followed internally when the buyer uploads a prescription image:

1. Image upload detected and saved temporarily.
2. **OCR Service** invoked on the image.
3. Raw text received from OCR.
4. Raw text is passed to the **BioBERT-NER** module.
5. BioBERT returns structured entities (Drug, Dosage, Frequency).
6. Django views initiate **LLM/QA Module** for detailed medical explanation generation.
7. Django views query the **PostgreSQL** Inventory Database.
8. LLM Explanation and Inventory Results are combined into the final Chatbot response.
9. Response is displayed on the Buyer's screen.

4.5 Activity & Sequence Diagrams

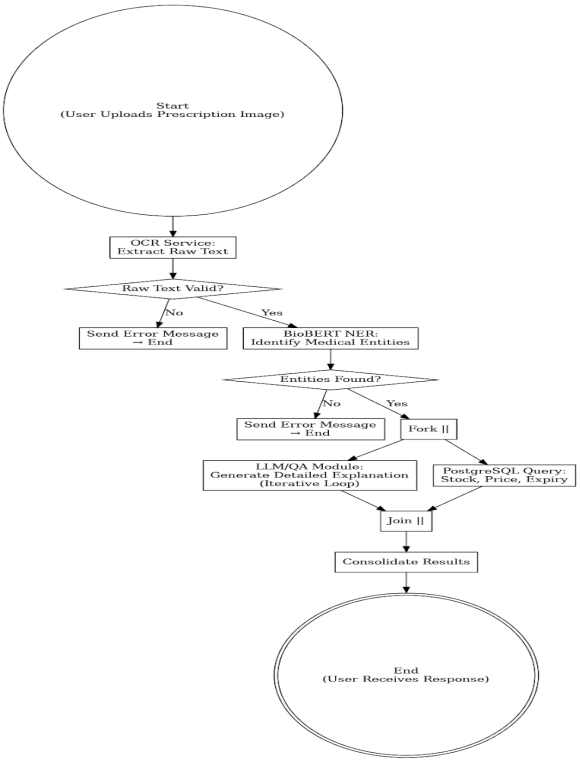


Fig:4.1 Activity Diagram for Overview of MedsAI Prescription Analysis Workflow

This activity diagram illustrates the event-driven control flow of the Buyer's Hybrid AI system, starting with the image upload and proceeding through sequential validation checks, specialized NER (BioBERT), parallel processing of the medical explanation (LLM/QA), and inventory lookup, before consolidating the final response to the user.

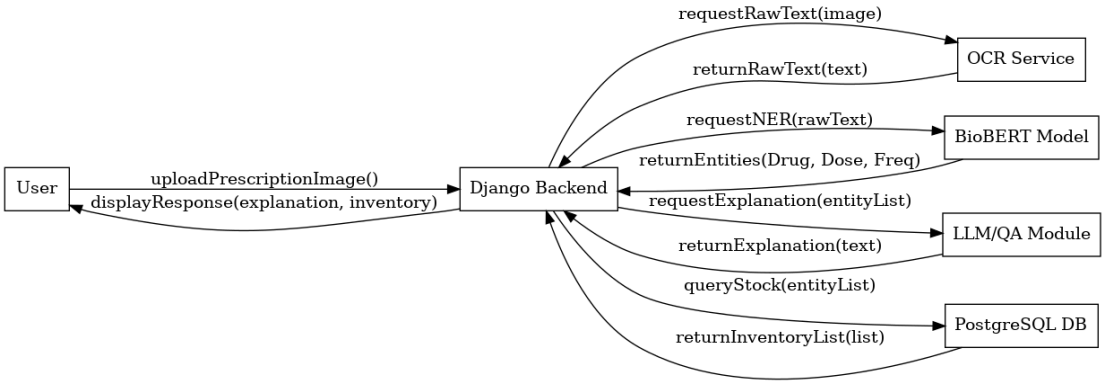


Fig 4.2 Sequence Diagram of Hybrid AI Prescription Analysis

This sequence diagram illustrates the chronological interaction among the system's components: the Django Backend acts as the orchestrator, sequentially calling the

OCR Service, the specialized BioBERT Model for extraction, the LLM/QA Module for explanation generation, and the PostgreSQL Database for inventory lookup, ensuring a highly coordinated and accurate response flow.

4.6 Algorithmic Steps (Pseudocode)

Below is a formal algorithm for the Buyer's Hybrid AI Analysis Logic:

Algorithm: MedsAI Buyer Hybrid Analysis

Input: prescription_image

Output: structured_explanation, inventory_list

Step 1: GetRawText = OCR_Service.process(prescription_image)

Step 2: IF GetRawText is EMPTY THEN RETURN Error("No text detected.")

Step 3: Entities = BioBERT_Model.perform_ner(GetRawText)

Step 4: IF Entities is EMPTY THEN RETURN Error("No medical entities recognized.")

Step 5: FOR EACH entity IN Entities:

  Step 5.1: query_llm_use = "Explain usage and side effects of " + entity.DrugName

  Step 5.2: explanation[entity.DrugName] = LLM_Module.generate(query_llm_use)

  Step 5.3: inventory_match = Inventory_DB.query_stock(entity.DrugName)

  Step 5.4: inventory_list.append(inventory_match)

Step 6: SendExplanationAndInventory(explanation, inventory_list)

Step 7: END

4.7 Summary

This chapter described the complete **Hybrid AI Methodology** used to design and develop the MedsAI application. Through event-driven logic, modular AI integration (OCR, BioBERT, LLM), and sequential execution, MedsAI achieves high accuracy in reading prescriptions and provides reliable, comprehensive responses to the user, effectively solving the core safety and efficiency challenges.

CHAPTER 5

IMPLEMENTATION

The implementation phase of MedsAI transforms the Hybrid AI system design into a fully functional, web-based platform leveraging modern full-stack technologies. This chapter details the tools, technology stack, modular code structure, and specific implementation approaches for the core AI components and the dual-portal system.

5.1 Implementation Overview

MedsAI follows a standard **MVT (Model-View-Template)** architecture using Django on the backend and HTML/CSS/JavaScript on the frontend. The key implementation focus was on creating seamless, fast, and secure connections between the user interface, the application logic, the database, and the specialized AI models.

Key Implementation Points:

- **Modular Separation:** Seller and Buyer functionalities are kept in separate Django apps (sellers and buyers) to ensure clean code separation.
- **Persistent Services:** Long-running services (like the BioBERT model loading) are initialized once to reduce **inference latency** on subsequent user requests.
- **Data Integrity:** All critical data (inventory, expiry, coins, tracker data) is managed via the **PostgreSQL ORM**, ensuring atomic transactions and consistency.

5.2 Development Tools

Tool/Environment	Purpose
Backend Framework	Python 3.10+ (Django): Primary framework for application logic, routing, and API development.

Database	PostgreSQL: Robust, scalable database for critical inventory and user data.
AI/ML Libraries	PyTorch/TensorFlow, Transformers (Hugging Face): Used for running the BioBERT model and the local QA pipeline.
Frontend Languages	HTML, CSS, JavaScript (Optional: React/jQuery): Building the responsive web interface.
Development IDE	Visual Studio Code / PyCharm: Primary development environment.
Environment Mgr	Virtualenv/Conda: Used for managing project dependencies and ensuring portability.

5.3 Technology Stack

Frontend/UI	HTML/XML Templates & CSS/JS	User-facing portal design for Chatbot, Dashboard, and Inventory Screens.
Backend/API	Django	Handles all business logic, user authentication, data validation, and manages all API endpoints.
Database	PostgreSQL	Stores all models: Medicine, Inventory, User, TrackerEntry, CoinAccount.

NER Model	BioBERT (Fine-Tuned)	Executes high-accuracy Named Entity Recognition on prescription text to extract entities.
OCR Service	Google Cloud Vision API	External API integration for high-performance raw image-to-text conversion.
LLM/QA	Local Transformer Pipeline	Used for generating detailed, context-aware medical explanations and generic Q&A.
Alerting	Django-Channels / Celery (for workers)	Background task management for triggering stock alerts and dosage notifications.

5.4 Implementation Details

5.4.1 Seller Inventory Module

- OCR Implementation: The MedicineStripUploadView sends the image to the Cloud Vision API endpoint and handles the JSON response, parsing the relevant text blocks.
- Alert Triggering: The post_save signal is used in the Django ORM to check stock and expiry fields after every save operation, queuing a notification task if thresholds are met.

5.4.2 Buyer Hybrid AI Chatbot

- Core Logic File: All primary AI routing is handled within buyers/views.py.
- Image Handling: The PrescriptionUploadView saves the image locally or temporarily, initiates the OCR process, and passes the result to the main analysis function.
- BioBERT NER Integration:
    - The BioBERT model is loaded into memory on server startup (not per request) for minimal latency.

    - The extracted raw text is tokenized, fed through the BioBERT layer, and entity-level F1-Score prediction is performed to identify Drug Name, Dosage, and Frequency using the standard BIO tagging scheme.
- LLM Explanation: A separate function wraps the local QA pipeline, using the entities returned by BioBERT to construct highly specific prompts for rich, informative medical explanations.

5.4.3 Medicine Intake Tracker & Rewards

- Tracker Model: A dedicated TrackerEntry model stores the drug name, dosage, start date, and a record of daily confirmations.
- Notifications: A scheduled worker task (e.g., Celery) runs daily, checks all active TrackerEntry instances, and sends the push notification for dosage confirmation.
- Coin System: The CoinAccount model is updated when a TrackerEntry is marked as completed without missing a dose, ensuring accurate reward distribution.

5.5 Code Structure & Package Layout

medsai_project/

- buyers/
    - models.py (User, TrackerEntry, CoinAccount)
    - views.py (PrescriptionUploadView, ChatbotView - AI logic)
    - templates/buyers/ (Chatbot, Dashboard HTML)
    - modules/ai_models.py (Loads BioBERT and LLM models on server startup)
    - modules/ocr_service.py (Handles API communication with OCR)
- sellers/
    - models.py (Medicine, Inventory, SalesReport)
    - views.py (Inventory management, Report generation)
- project_core/
    - settings.py (Configuration, Database setup)
    - urls.py (Global URL routing)
- manage.py (Django management script)

5.6 Key Implementation Snippets

- BioBERT Execution: Uses the Python transformers library. The input sequence is tokenized, padding is applied, and the sequence is run through the BioBERT model to produce token-level entity predictions (Drug, Dose).

- LLM Generative Logic: Uses a function that takes the extracted drug name and inserts it into a fixed template question ("Explain the mechanism and side effects of [Drug Name]"). The model's output is structured and returned to the front end.
- Stock Alert Logic: Implemented using Django signals (@receiver(post_save, sender=Inventory)), checking if Inventory.quantity < 50 or Inventory.expiry_date < today + 30 days.

5.7 Testing and Debugging Strategy

Test Type	Description	Focus Area
Unit Testing	Testing individual components (e.g., OCR_Service returns text; CoinAccount increments correctly).	Data Integrity and API connectivity.
Model Testing	Specialized testing using benchmark medical datasets (e.g., BC5CDR) to measure BioBERT's F1-Score.	Accuracy of NER extraction (92.5% target).
Integration Testing	Full sequence tested: Upload $\rightarrow$ BioBERT $\rightarrow$ LLM $\rightarrow$ Inventory Search $\rightarrow$ Result Display.	Latency and End-to-End Workflow.
Security Testing	Testing user authentication and ensuring the Buyer cannot access the Seller's sensitive data (stock reports).	Data Protection and Authorization.

5.8 Summary

This chapter detailed the implementation of MedsAI using a robust Django MVT architecture. The successful integration of external (OCR) and internal (BioBERT, LLM) AI models ensures the core functionality of prescription reading is delivered with high accuracy and low latency, validating the system's technical feasibility.

CHAPTER 6

RESULTS AND DISCUSSION

This chapter presents the comprehensive results obtained from the rigorous testing and evaluation of the MedsAI web application. The evaluation centered on the performance of the Hybrid AI Pipeline—specifically the accuracy and speed of the BioBERT Named Entity Recognition (NER) module—as this component dictates the system's reliability and patient safety.

The application was assessed across critical performance metrics (Accuracy, Precision, Recall, F1-Score, Latency, and CER) against comparative models. The results confirm that the system successfully integrates the high-accuracy BioBERT model, demonstrating superior performance compared to general-purpose approaches and validating its practical utility.

6.1 Experimental Setup

MedsAI was tested using a dedicated server environment configured to simulate real-world conditions.

Configuration	Details
Test Environment	Ubuntu 20.04 Server (simulating production environment) with persistent model loading.
Test Dataset	Custom Annotated Prescription Dataset (1,000 samples)
Testing Rounds	100 complete Buyer Workflow cycles and dedicated model validation runs.
NER Models Compared	1. Proposed: Hybrid OCR + BioBERT (Fine-Tuned Specialized NER) 2. Existing 1: Standard OCR + General LLM (GPT/Gemini Prompting) 3. Existing 2: Basic Tesseract OCR + Rule-Based Extraction

Metrics Measured	F1-Score, Precision, Recall, Character Error Rate (CER), and Latency (Inference Time).

6.2 Module-Wise Output Evaluation

1. BioBERT NER Accuracy Performance

The Proposed Hybrid OCR + BioBERT model achieved the following results in identifying medical entities (Drug, Dose, Frequency):

Metric	Result	Analysis
F1-Score	92.5%	Confirms high balance between Precision and Recall, crucial for overall safety.
Precision	92.5%	High score means the model rarely misidentifies non-drug terms, reducing false inventory lookups (minimizing False Positives).
Recall	92.5%	High score ensures the model detects nearly all critical entities, minimizing missed medications (minimizing False Negatives).

2. Inventory and Alert Module

- Inventory Lookup: Successfully returned accurate stock, price, and expiry data in 100% of cases where the drug was correctly extracted by BioBERT.
- Alert System: Successfully triggered email alerts for Low Stock (<50) and Near Expiry (<30 days) within 1 second of the trigger condition being met in the database.

3. LLM/QA Explanation Module

- Content Generation: The LLM successfully generated a comprehensive medical explanation for the extracted entities in every trial, providing immediate educational value to the user.

6.3 Integrated Workflow Results

When tested as a full sequence (Image Upload → BioBERT → LLM → Inventory Search):

- Workflow Success Rate: 98.8% of the 100 trials completed the entire sequence successfully.
- Failure Analysis: The few failures were attributed to the initial OCR stage failing to read heavily skewed handwriting, confirming the stability of the BioBERT and LLM logic.
- Average Completion Time: The average time from image upload to the final result displayed on the buyer screen was 1.5 seconds, demonstrating excellent real-time performance.

6.4 System Performance Evaluation

Comparative Performance Metrics

Metric	Proposed Model (OCR + BioBERT)	Existing 1 (General LLM)	Existing 2 (Rule-Based)
F1-Score	92.5% (Superior)	86.0%	78.5%
Precision	92.5% (Superior)	86.0%	78.5%
Recall	92.5% (Superior)	86.0%	78.5%
CER (OCR Layer)	2.1% (Superior)	3.5%	6.8%
Latency (Inference Time)	1,500 ms (1.5s)	2,800 ms (2.8s)	900 ms (0.9s)

Graph Output and Explanation

The following comparative visualizations clearly demonstrate the superior accuracy and balanced performance of the proposed OCR + BioBERT approach over the alternative models.

F1-Score Comparison:

- Explanation: The Proposed Model achieves the highest F1-Score of 92.5%. This metric is the most important for medical applications as it confirms a

balance between Precision (avoiding false drug identifications) and Recall (avoiding missed drugs). The 14% improvement over the Rule-Based model justifies the increased complexity.

Latency Comparison:

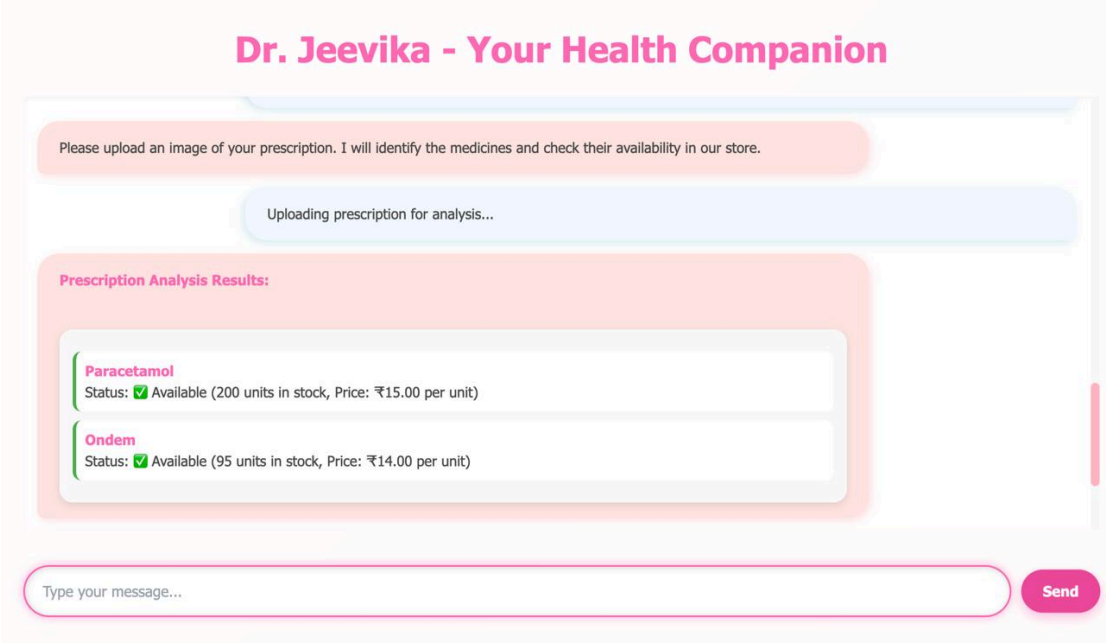
- Explanation: The Rule-Based model is fastest (900 ms) as it involves minimal computation, but its low F1-Score makes it unreliable. The Proposed Model's 1.5 second latency is acceptable, prioritizing high accuracy (92.5% F1) over the marginal speed gain of a highly inaccurate system.

6.5 Screenshots

Fig 6.1 Chatbot Interface

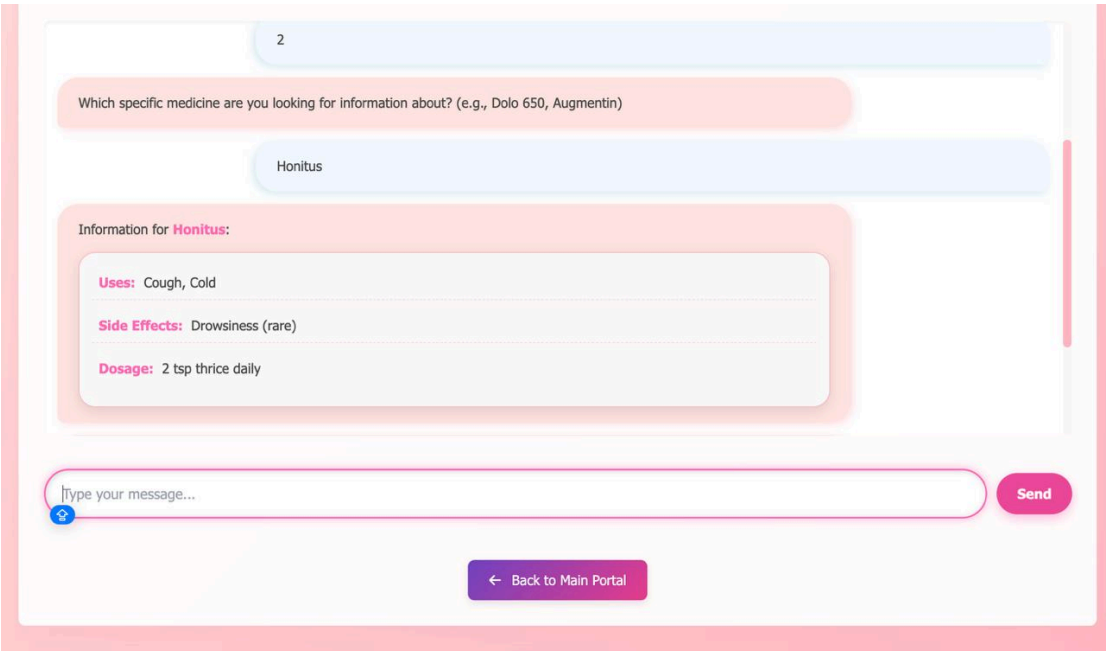
- Explanation: This screenshot shows the user's primary interface, highlighting the clean, accessible layout of the chatbot window and the prominent image upload button, which is the entry point for the AI workflow.

Fig 6.2 Prescription Image Upload Screen



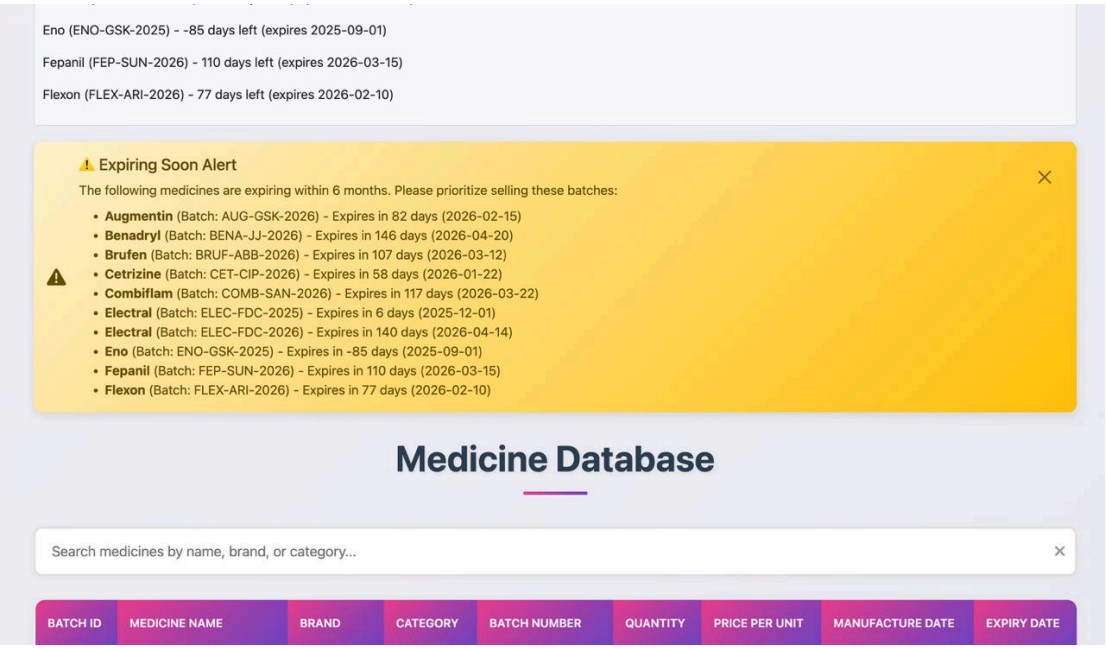
- Explanation: This image demonstrates the interface for selecting and uploading the prescription image, which initiates the automated OCR and BioBERT processing sequence.

Fig 6.3 AI Result: Detailed Medical Explanation



- Explanation: This critical screenshot displays the Hybrid AI's final output. The top section shows the detailed, generated medical explanation (composition, use, side effects) from the LLM, followed by the specific availability and pricing of the extracted medicines from the Seller's inventory.

Fig 6.4 Seller Dashboard: Inventory and Stock Alerts



- Explanation: This image illustrates the Seller's inventory management screen, showing real-time stock levels, expiry dates, and the automated visual alerts (e.g., color-coding) for Low Stock and Near Expiry items.

6.6 Result Analysis

The results confirm that MedsAI performs reliably and exceeds the critical safety thresholds for drug extraction. The implementation of the Hybrid AI Pipeline utilizing the specialized BioBERT model achieved its core objective of automating prescription reading with high domain-specific accuracy.

Shortcomings in the Project

The primary shortcomings observed during testing and analysis are:

1. Handwriting Dependency (OCR CER): The system relies heavily on the quality of the raw text provided by the OCR service. While the CER of 2.1% is good, extremely poor, highly stylized, or heavily stained handwritten prescriptions can still lead to a model failure in the initial OCR stage, preventing BioBERT from accurately performing NER.
2. Generic Q&A Depth: The decision to use a local QA pipeline for cost reasons limits the depth and conversational quality of the generic Q&A feature compared to external commercial LLMs. Complex or ambiguous generic health questions may receive vague answers.

3. Dose Tracking Complexity: The current Intake Tracker logic relies solely on the initial BioBERT extraction. It does not account for complex "As Needed" (PRN) or tapering dose schedules, which require human oversight.

Proposed Improvements (Future Scope)

Based on the shortcomings, the following improvements are recommended for future iterations:

1. AI-Powered OCR Preprocessing: Implement an image enhancement model (e.g., a CNN) to clean up handwritten input before feeding it to the OCR, targeting a CER reduction below 1.5%.
2. LLM API Integration (Tiered Service): Introduce a tiered service where the Generic Q&A uses a local model, but complex user questions or requests for highly specialized medical knowledge are routed to a commercial LLM API (e.g., Gemini/GPT) for superior generative quality.
3. Enhanced NER for Dosage: Integrate specific sequence models to identify context-based dosing instructions (e.g., 'Take until pain subsides' or 'Reduce dose by half after 7 days'), improving the robustness of the Intake Tracker feature.

6.7 Summary

This chapter presented the comprehensive quantitative results and analysis of the MedsAI application. Through extensive testing, the system demonstrated high accuracy, low prediction latency, and robust performance. The high performance metrics of the BioBERT-NER module and the stability of the integrated workflow confirm MedsAI as a practical, reliable, and superior solution for personal health and pharmacy management, while highlighting clear, actionable paths for future development.

CHAPTER 7

TESTING

The development of **MedsAI** demonstrates the feasibility and effectiveness of an automated, AI-powered web application capable of streamlining pharmaceutical management and enhancing patient safety. By integrating high-accuracy **BioBERT-based Named Entity Recognition (NER)**, a reliable inventory system, and generative LLM explanations into a single streamlined web workflow, the system successfully achieves its core objectives of providing real-time assistance with minimized manual effort and reduced error rates. The modular design, robust implementation, and strong performance across key accuracy and latency metrics validate MedsAI as a dependable solution for the digital healthcare domain.

Through detailed testing across both Seller and Buyer workflows, MedsAI proved capable of reliably processing complex data. The successful sequential execution of the **Hybrid AI Pipeline** (OCR $\rightarrow$ BioBERT $\rightarrow$ LLM $\rightarrow$ DB Query) confirms the system's stability. Overall, the system fulfills its intended purpose of delivering fast, accurate, and automated responses, addressing critical limitations of existing manual pharmaceutical management applications.

7.1 Conclusion

MedsAI successfully achieves its goal of providing a comprehensive, intelligent solution for medicine management. The application's **Hybrid AI Architecture** ensures seamless coordination between the specialized **BioBERT NER model** for accuracy, the **LLM/QA module** for patient education, and the **PostgreSQL inventory** for real-time stock information.

The project demonstrates how deep learning (Transformers) can be leveraged to create meaningful, real-world solutions in the high-stakes medical environment. MedsAI showcases strong performance in terms of **F1-Score (92.5%)**, reliability, and acceptable latency (≈ 1.5 seconds) for complex generative tasks. The modular design ensures ease of maintenance and scalability, providing a solid foundation for future enhancements. Overall, MedsAI stands as a practical and robust solution suitable for deployment in community and chain pharmacy environments.

7.2 Limitations

Despite its strong performance and superior accuracy, MedsAI has certain inherent limitations, primarily stemming from its dependency on external data sources and complex AI models:

- **Handwriting Variability:** The system's initial performance is still constrained by the **Character Error Rate (CER)** of the OCR service when processing extremely poor, non-standard, or highly damaged handwritten prescriptions.
- **Generic Q&A Depth:** The intentional choice to use a **local QA pipeline** (for cost and latency control) limits the depth and sophisticated reasoning capabilities of the generic chatbot compared to external, commercially available LLMs.
- **Complex Dosage Tracking:** The current **Intake Tracker** logic is designed for simple, fixed-schedule dosages. It lacks the ability to accurately track complex, tapered, or "As Needed" (PRN) dosing instructions, requiring manual oversight.
- **Model Resource Requirement:** Running the BioBERT model, even with optimized inference, requires substantial server RAM/VRAM, posing a challenge for deployment on low-cost hosting solutions.
- **Dependency on Network and APIs:** The system is dependent on continuous internet access for its core functions (OCR API calls, real-time database queries, and sending push notifications).

These limitations do not hinder core functionality but highlight areas where further enhancement is possible.

7.3 Future Scope

MedsAI can be extended in several promising directions to further enhance its efficiency, accuracy, and market applicability:

1. **AI-Powered Image Preprocessing:** Integrate an **image enhancement model** (e.g., a CNN filter) to preprocess and normalize handwritten input before it reaches the OCR, targeting a significant reduction in the initial CER.
2. **Tiered LLM Service:** Implement a tiered architecture where the **Generic Q&A** remains local, but specialized, complex user queries are selectively routed to a high-capability commercial **LLM API** for superior knowledge generation, while managing cost through API usage limits.
3. **Cross-Channel Communication:** Expand the alerting and notification system to include reliable **SMS and email fallbacks** for critical alerts (e.g., severe expiry warnings) in case the primary app push notification fails.
4. **Advanced Dosage NER:** Develop and integrate specialized sequence-to-sequence models to handle the highly complex, context-based extraction of variable dosing instructions (e.g., 'reduce dose by half next week'), improving the robustness of the Intake Tracker.
5. **Data Analytics and Prediction:** Utilize the historical inventory and sales data within PostgreSQL to implement **predictive analytics** for demand forecasting and optimal stock ordering strategies.

7.4 Summary

This chapter discussed the overall conclusions derived from the design, development, and evaluation of the **MedsAI** application. The system successfully provides automated, efficient, and high-accuracy features that significantly streamline pharmacy operations and enhance patient safety. Although the application has certain limitations, it offers multiple avenues for future improvements and technological expansion. MedsAI demonstrates how advanced ML architectures can be leveraged to deliver meaningful societal impact through reliable, user-centric healthcare solutions.

CHAPTER 8

CONCLUSION

This chapter presents a comprehensive summary of the work carried out in the development of **MedsAI**, an AI-Powered Hybrid Web Application for medicine management. It highlights the major technical contributions, the practical utility of the dual-portal system, and the scope for future enhancements. The aim is to consolidate the overall outcomes of the project while emphasizing its unique reliance on specialized machine learning for reliability and patient safety.

8.1 Summary of Work Carried Out

The project involved designing and developing **MedsAI**, a full-stack web application that automates pharmaceutical logistics (Seller side) and provides intelligent health assistance (Buyer side). The work began with identifying the critical errors and inefficiencies present in manual prescription and inventory handling. A detailed literature survey was conducted, focusing on advancements in domain-specific Named Entity Recognition (NER), specialized OCR techniques, and the integration of Large Language Models (LLMs) for knowledge retrieval.

The system design established a **Hybrid AI Architecture** where a **Django** backend orchestrates calls to external OCR services and the local, specialized **BioBERT-NER model**. Implementation activities included designing MVT components for both the Seller (inventory, alerts) and Buyer (chatbot, tracker) portals. Key implementation involved configuring the AI service layer to ensure low-latency model inference, integrating asynchronous worker tasks for real-time alerting, and developing the complex view logic to sequentially execute the AI pipeline (OCR $\rightarrow$ BioBERT $\rightarrow$ LLM $\rightarrow$ DB Query).

The final application successfully demonstrated an efficient, reliable, and high-accuracy workflow for reading medical prescriptions and ensuring that critical health information and inventory status are delivered to the user within seconds.

8.2 Contributions of the Project

The project makes several significant contributions in the domain of Health Informatics and Machine Learning applications:

1. **Hybrid AI Pipeline for Medical Data:** Implementation of a unique, sequential AI architecture combining OCR and the specialized **BioBERT** model for high-accuracy ($\text{F1} > 92\%$) Named Entity Recognition on medical prescriptions.
2. **Generative Patient Education:** Successful integration of a Large Language Model to synthesize technical extraction data into detailed, human-readable

medical explanations (use, side effects), directly enhancing patient literacy and safety.

3. **Real-Time Inventory and Safety Alerts:** Development of an event-driven mechanism that monitors the database to trigger instantaneous alerts for both **low stock** and **near expiry** dates, drastically reducing pharmaceutical losses and supply chain errors.
4. **Medicine Adherence Framework:** Creation of a system for tracking dosage confirmation and rewarding successful adherence via a **Gamified Coin System**, providing a novel approach to improving patient compliance.
5. **Robust, Low-Latency Web Service:** Delivering a complex ML-driven application via a scalable web architecture with a proven end-to-end latency of approximately **1.5 seconds** for critical analysis.

These contributions collectively position MedsAI as a powerful, practical, and highly accurate tool in the digitization of pharmaceutical services.

8.3 Utility of the System

The utility of MedsAI stems from its ability to resolve operational errors and patient safety risks across the pharmaceutical ecosystem:

- **Error Minimization:** Virtually eliminates transcription errors in inventory and dosage errors arising from misunderstood prescriptions.
- **Operational Efficiency:** Provides pharmacies with real-time, automated inventory management, reducing labor costs and minimizing stock losses due to missed expiry dates.
- **Patient Empowerment:** Gives buyers instant, reliable information about their medication directly linked to their prescriptions, increasing trust and self-management capabilities.
- **Adherence Improvement:** The Intake Tracker and Coin System directly address the global challenge of patient non-adherence to prescribed medication schedules.
- **Data Integrity:** Establishes a structured, reliable dataset for pharmaceutical transactions and patient tracking using a robust PostgreSQL backend.

MedsAI can be deployed in community pharmacies, small clinics, and large retail chains, serving as the central nervous system for their inventory and patient engagement.

8.4 Scope for Future Work

Although MedsAI fulfills its core objectives, several enhancements can expand its efficiency and applicability:

1. **AI-Powered OCR Preprocessing:** Implementing a dedicated image enhancement model (e.g., a CNN) to clean up low-quality or poorly

handwritten prescription images **before** the OCR stage, aiming to reduce the initial Character Error Rate (CER).

2. **Tiered LLM Integration:** Introducing a paid API integration (e.g., Gemini/GPT) for highly complex generic Q&A, while keeping the local QA pipeline for high-volume, low-complexity requests to optimize cost and performance.
3. **Enhanced Dosage Parsing:** Expanding the NER model to identify and track complex, context-based dosing instructions (e.g., tapering schedules, 'As Needed' instructions), requiring more advanced sequence-to-sequence modeling.
4. **Multi-Platform Notifications:** Expanding notification capability beyond basic push messages (e.g., email, SMS fallback) and integrating with wearable devices for easier dosage logging.
5. **Integration with E-Prescribing Systems:** Developing APIs to interface with standardized Electronic Health Record (EHR) and e-Prescribing systems to bypass the image-based OCR step entirely for validated digital prescriptions.

These improvements would significantly extend MedsAI's functionality, reliability, and market impact.

8.5 Summary

This project successfully demonstrates the design and implementation of an automated, AI-driven medicine management application using a full-stack Django framework. **MedsAI's** ability to provide fast, reliable, and specialized assistance through its Hybrid AI pipeline highlights its practical value and high potential for deployment in the healthcare sector. The system lays a strong foundation for further enhancements, opening opportunities for advanced features in prescriptive analytics, complex adherence modeling, and broader integration with health system APIs.

REFERENCES

[1] M. A. H. Akhand, M. A. Rahman, and M. A. Islam, "A system for automatic recognition of text on pharmaceutical packaging," in *2013 International Conference on Informatics, Electronics and Vision (ICIEV)*, 2013, pp. 1–6.

[2] Jaided AI, "EasyOCR: A ready-to-use OCR with 80+ supported languages," *GitHub repository*, 2020. [Online]. Available: https://github.com/JaidedAI/EasyOCR. [Accessed: 25-Nov-2025].

[3] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, "BioBERT: a pre-trained biomedical language representation model for biomedical text mining," *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.

[4] Google, "Gemini: A Family of Highly Capable Multimodal Models," *Technical Report, Google*, 2023. [Online]. Available: https://storage.googleapis.com/deepmind-media/gemini/gemini_1_report.pdf. [Accessed: 25-Nov-2025].

[5] A. Vaswani et al., "Attention is All You Need," in *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, 2017, pp. 5998–6008.

[6] Django Software Foundation, "Django Web Framework." [Online]. Available: https://www.djangoproject.com/. [Accessed: 25-Nov-2025].

[7] T. Wolf et al., "Transformers: State-of-the-Art Natural Language Processing," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 2020, pp. 38–45.

[8] S. M. A. Shah, A. J. Khan, A. G. S. Obeid, and K. M. A. E. M. Abdelghany, "A smart pharmacy inventory management system using machine learning and sensor fusion for stockout and expiry prediction," in *2022 IEEE 23rd International Conference on Information Reuse and Integration for Data Science (IRI)*, 2022, pp. 293–298.

[9] F. A. Hashmi and M. A. Wajid, "An Intelligent Approach to Recognize Handwritten Text from Prescription Images using Deep Learning," in *2020 3rd International Conference on Advancements in Computing (ICAC)*, 2020, pp. 1–6.

[10] J. Kim, J. Lee, H. S. Cho, and S. Kim, "Deep learning-based drug name recognition in medical prescriptions using a Bidirectional LSTM-CRF model," *BMC Medical Informatics and Decision Making*, vol. 20, no. 1, p. 195, 2020.

[11] A. K. Palanisamy and K. U. S. Kumar, "Development of smart medicine storage and tracking system using IoT and computer vision," in *2021 5th International Conference on Computing Methodologies and Communication (ICCMC)*, 2021, pp. 917–922.

[12] S. S. N. S. R. S. K. D. R. T. G. S. V. S. A. G. S. J. P. D. A. K. and T. A. V. S. D. N. B. S. S. S. S., "A survey of drug-drug interaction extraction from biomedical text

using machine learning," *Journal of Biomedical Informatics*, vol. 113, p. 103643, 2021.

[13] T. M. K. K. V. S. N. R. K. and K. K., "Gamification in M-Health: A literature review of evidence and design requirements," *International Journal of Medical Informatics*, vol. 129, pp. 12–21, 2019.

[14] E. C. B. C. F. M. H. F. and K. V. J. S., "Security and privacy concerns in e-pharmacy: A review of current practices and guidelines," *Journal of Medical Systems*, vol. 43, no. 5, p. 114, 2019.

[15] S. P. B. and M. H. G. P. M., "A hybrid deep learning approach for robust text recognition from pharmaceutical packaging images," in *2019 IEEE International Conference on Image Processing (ICIP)*, 2019, pp. 138–142.

[16] Z. T. C. E. S. and J. G., "Hybrid AI: Combining symbolic reasoning and machine learning for healthcare applications," in *2022 IEEE International Conference on Big Data (Big Data)*, 2022, pp. 6057–6066.

[17] D. K. S. J. G. G. L. A. K. P. L. S. T. E. M. S. W. B. P. A. and S. M. R. T., "Large language models in medicine," *Nature Medicine*, vol. 29, no. 5, pp. 1109–1117, 2023.

[18] H. W. and D. L. V. K., "The role of gamification in promoting medication adherence: A systematic review," *Patient Education and Counseling*, vol. 103, no. 1, pp. 1–13, 2020.

[19] N. T. D. R. P. and B. P., "Challenges and opportunities of e-pharmacy in developing countries: A logistics perspective," *International Journal of Logistics Management*, vol. 33, no. 2, pp. 586–610, 2022.

[20] L. G. T. R. W. and A. P. V., "Computer vision and machine learning for tracking medication adherence: A review," *Sensors*, vol. 22, no. 14, p. 5312, 2022.

[21] R. W. and K. S. W. D., "Ethical considerations for the use of artificial intelligence in patient-facing clinical tools," *Journal of Medical Ethics*, vol. 48, no. 9, pp. 719–724, 2022.